

Multiple Choice (10 marks)

1. When the Heisenberg uncertainty principle is applied to a quantum mechanical particle in the lowest energy level of a one-dimensional box, which of the following is true?
 - (a) Momentum is known exactly, but no information about position can be known.
 - (b) Position is known exactly, but no information about momentum can be known.
 - (c) No information about either position or momentum can be known.
 - (d) Neither position nor momentum can be known exactly.
2. Which of the following statements most accurately explains why the ^1H spectrum of $^{12}\text{CHCl}_3$ is a singlet?
 - (a) Both ^{35}Cl and ^{37}Cl have $I = 0$.
 - (b) The hydrogen atom undergoes rapid intermolecular exchange.
 - (c) The molecule is not rigid.
 - (d) Both ^{35}Cl and ^{37}Cl have electric quadrupole moments.
3. Which one of the following statements is true:
 - (a) Protons and neutrons have orbital and spin angular momentum.
 - (b) Protons have orbital and spin angular momentum, neutrons have spin angular momentum.
 - (c) Protons and neutrons possess orbital angular momentum only.
 - (d) Protons and neutrons possess spin angular momentum only.
4. Which of the following is classified as a conjugate acid-base pair?
 - (a) HCl / NaOH
 - (b) $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
 - (c) $\text{O}_2 / \text{H}_2\text{O}$
 - (d) H^+ / Cl^-
5. Which of the following statements about the lanthanide elements is NOT true?
 - (a) The most common oxidation state for the lanthanide elements is +3.
 - (b) Lanthanide complexes often have high coordination numbers (> 6).
 - (c) All of the lanthanide elements react with aqueous acid to liberate hydrogen.
 - (d) The atomic radii of the lanthanide elements increase across the period from La to Lu.

True / False (10 marks)

Answer True or False

6. True or False: CH_3Cl has a ^1H resonance approximately 1.55 kHz downfield from TMS in a 12.0 T magnetic field.
7. True or False: A magnetic field of 1.2 T is sufficient to achieve the polarization of 2.5×10^{-6} for ^{13}C at 298 K.
7. True or False: Unpaired electrons are essential for both paramagnetism and ferromagnetism in materials.
8. True or False: Carbonic acid is not generally considered a toxic pollutant in water systems.
9. True or False: Ammonia (NH_3) is the strongest base in liquid ammonia.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the orbital angular momentum quantum number, l , of the electron that is most easily removed from ground-state aluminum during ionization. What factors influence this?
12. Discuss the factors that contribute to the difference in first ionization energy between argon and neon, and explain why argon's ionization energy is lower.

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